Atoms, Ions and Reactions: Notation for Chemistry and Nuclear Science

Formulas and Ions

Chemical formulas are set upright, with subscripts for the number of atoms and superscripts for charge. The formula of water is H_2O. The formula of calcium hydroxide is $Ca(OH)_2$. The formula of glucose is $C_6H_{12}O_6$. The formula of ammonia is NH_3. The formula of sulfuric acid is H_2SO_4. The formula of ethanol is C_2H_6O. The formula of aluminium sulfate is $Al_2(SO_4)_3$. The formula of sodium bicarbonate is $NaHCO_3$. The formula of methane is CH_4. The formula of iron(III) oxide is Fe_2O_3. The sodium cation is written Na^+ and carries a charge of 1. The chloride anion is written Cl^- and carries a charge of 1. The calcium cation is written Ca^{2+} and carries a charge of 2. The sulfate anion is written SO_4^{2-} and carries a charge of 2. The iron(III) cation is written Fe^{3+} and carries a charge of 3. The ammonium cation is written NH_4^+ and carries a charge of 1. The hydroxide anion is written OH^- and carries a charge of 1. The phosphate anion is written PO_4^{3-} and carries a charge of 3. The copper(II) cation is written Cu^{2+} and carries a charge of 2. The nitrate anion is written NO_3^- and carries a charge of 1. Aluminium sulfate, $Al_2(SO_4)_3$, is made of molecules or ions that differ from those of ammonia, NH_3. The copper(II) ion, Cu^{2+}, is formed when the neutral species loses 2 electrons. When the salt of calcium and phosphate dissolves in water it releases Ca^{2+} and $PO_4{}^{3-}$ ions. A solution of methane (CH_4) was mixed with a solution containing $NH_4{}^+$ ions.

In the reaction NH_3 reacts with H_2SO_4 to give H_2O, the mass of the products equals the mass of the reactants. Water has the formula H_2O and is stored in a sealed bottle. A bottle labelled chloride holds the ion Cl^-, which is surrounded by water molecules. A bottle labelled nitrate holds the ion $NO_3{}^-$, which is surrounded by water molecules.

In the reaction C_2H_6O reacts with $Al_2(SO_4)_3$ to give CH_4, the mass of the products equals the mass of the reactants. Sodium bicarbonate, $NaHCO_3$, is made of molecules or ions that differ from those of sulfuric acid, H_2SO_4. A solution of aluminium sulfate ($Al_2(SO_4)_3$) was mixed with a solution containing Fe^{3+} ions. When the salt of sodium and phosphate dissolves in water it releases Na^+ and $PO_4{}^{3-}$ ions.

Glucose has the formula $C_6H_{12}O_6$ and is stored in a sealed bottle. The calcium ion, Ca^{2+}, is formed when the neutral species loses 2 electrons. A solution of aluminium sulfate ($Al_2(SO_4)_3$) was mixed with a solution containing Fe^{3+} ions. In the reaction Fe_2O_3 reacts with $C_6H_{12}O_6$ to give $Ca(OH)_2$, the mass of the products equals the mass of the reactants.

When the salt of copper(II) and sulfate dissolves in water it releases Cu^{2+} and $SO_4{}^{2-}$ ions. The ammonium ion, $NH_4{}^+$, is formed when the neutral species loses one electron. Ammonia has the formula NH_3 and is stored in a sealed bottle. A bottle labelled phosphate holds the ion $PO_4{}^{3-}$, which is surrounded by water molecules.

Aluminium sulfate, $Al_2(SO_4)_3$, is made of molecules or ions that differ from those of ethanol, C_2H_6O. A solution of sodium bicarbonate ($NaHCO_3$) was mixed with a solution containing Cu^{2+} ions. The calcium ion, Ca^{2+}, is formed when the neutral species loses 2 electrons. In the reaction Fe_2O_3 reacts with H_2SO_4 to give $NaHCO_3$, the mass of the products equals the mass of the reactants.

A bottle labelled nitrate holds the ion $NO_3{}^-$, which is surrounded by water molecules. Iron(III)

oxide, Fe_2O_3, is made of molecules or ions that differ from those of ammonia, NH_3. When the salt of copper(II) and nitrate dissolves in water it releases Cu^{2+} and $NO_3{}^-$ ions. Aluminium sulfate has the formula $Al_2(SO_4)_3$ and is stored in a sealed bottle.

A bottle labelled nitrate holds the ion $NO_3{}^-$, which is surrounded by water molecules. A solution of sodium bicarbonate ($NaHCO_3$) was mixed with a solution containing Fe^{3+} ions. The ammonium ion, $NH_4{}^+$, is formed when the neutral species loses one electron. In the reaction CH_4 reacts with C_2H_6O to give Fe_2O_3, the mass of the products equals the mass of the reactants.

Chemical Equations

Each equation is balanced for atoms and, where ions appear, for charge. Mixing two clear solutions can form a solid. The net ionic equations below leave out the spectator ions:

$$Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$$

$$Pb^{2+}(aq) + 2\,I^-(aq) \longrightarrow PbI_2(s)$$

$$Ba^{2+}(aq) + SO_4^{2-}(aq) \longrightarrow BaSO_4(s)$$

$$Ca^{2+}(aq) + CO_3^{2-}(aq) \longrightarrow CaCO_3(s)$$

$$Fe^{3+}(aq) + 3\,OH^-(aq) \longrightarrow Fe(OH)_3(s)$$

$$Cu^{2+}(aq) + S^{2-}(aq) \longrightarrow CuS(s)$$

$$2\,Ag^+(aq) + CrO_4^{2-}(aq) \longrightarrow Ag_2CrO_4(s)$$

$$Mg^{2+}(aq) + 2\,OH^-(aq) \longrightarrow Mg(OH)_2(s)$$

$$3\,Ca^{2+}(aq) + 2\,PO_4^{3-}(aq) \longrightarrow Ca_3(PO_4)_2(s)$$

An acid and a base give a salt and water. The coefficients are chosen so that every atom is conserved:

$$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$$

$$H_2SO_4(aq) + 2\,KOH(aq) \longrightarrow K_2SO_4(aq) + 2\,H_2O(l)$$

$$2\,HNO_3(aq) + Ca(OH)_2(aq) \longrightarrow Ca(NO_3)_2(aq) + 2\,H_2O(l)$$

$$2\,H_3PO_4(aq) + 3\,Ba(OH)_2(aq) \longrightarrow Ba_3(PO_4)_2(s) + 6\,H_2O(l)$$

$$3\,\mathrm{HCl(aq)} + \mathrm{Al(OH)_3(aq)} \longrightarrow \mathrm{AlCl_3(aq)} + 3\,\mathrm{H_2O(l)}$$

Complete combustion of a fuel in oxygen produces carbon dioxide and water:

$$\mathrm{CH_4(g)} + 2\,\mathrm{O_2(g)} \longrightarrow \mathrm{CO_2(g)} + 2\,\mathrm{H_2O(l)}$$

$$\mathrm{C_3H_8(g)} + 5\,\mathrm{O_2(g)} \longrightarrow 3\,\mathrm{CO_2(g)} + 4\,\mathrm{H_2O(l)}$$

$$\mathrm{C_5H_{12}(l)} + 8\,\mathrm{O_2(g)} \longrightarrow 5\,\mathrm{CO_2(g)} + 6\,\mathrm{H_2O(l)}$$

$$\mathrm{C_2H_4(g)} + 3\,\mathrm{O_2(g)} \longrightarrow 2\,\mathrm{CO_2(g)} + 2\,\mathrm{H_2O(l)}$$

$$\mathrm{C_2H_6O(l)} + 3\,\mathrm{O_2(g)} \longrightarrow 2\,\mathrm{CO_2(g)} + 3\,\mathrm{H_2O(l)}$$

$$\mathrm{C_6H_{12}O_6(s)} + 6\,\mathrm{O_2(g)} \longrightarrow 6\,\mathrm{CO_2(g)} + 6\,\mathrm{H_2O(l)}$$

Active metals dissolve in acids and release hydrogen gas:

$$\mathrm{Mg(s)} + 2\,\mathrm{HCl(aq)} \longrightarrow \mathrm{MgCl_2(aq)} + \mathrm{H_2(g)}$$

$$\mathrm{Zn(s)} + 2\,\mathrm{HCl(aq)} \longrightarrow \mathrm{ZnCl_2(aq)} + \mathrm{H_2(g)}$$

$$2\,\mathrm{Al(s)} + 3\,\mathrm{H_2SO_4(aq)} \longrightarrow \mathrm{Al_2(SO_4)_3(aq)} + 3\,\mathrm{H_2(g)}$$

$$\mathrm{Fe(s)} + 2\,\mathrm{HCl(aq)} \longrightarrow \mathrm{FeCl_2(aq)} + \mathrm{H_2(g)}$$

A more reactive metal displaces a less reactive one from solution:

$$\mathrm{Zn(s)} + \mathrm{Cu^{2+}(aq)} \longrightarrow \mathrm{Zn^{2+}(aq)} + \mathrm{Cu(s)}$$

$$\mathrm{Mg(s)} + \mathrm{Fe^{2+}(aq)} \longrightarrow \mathrm{Mg^{2+}(aq)} + \mathrm{Fe(s)}$$

$$2\,\mathrm{Al(s)} + 3\,\mathrm{Cu^{2+}(aq)} \longrightarrow 2\,\mathrm{Al^{3+}(aq)} + 3\,\mathrm{Cu(s)}$$

$$\mathrm{Cu(s)} + 2\,\mathrm{Ag^{+}(aq)} \longrightarrow \mathrm{Cu^{2+}(aq)} + 2\,\mathrm{Ag(s)}$$

$$\mathrm{Zn(s)} + 2\,\mathrm{Ag^{+}(aq)} \longrightarrow \mathrm{Zn^{2+}(aq)} + 2\,\mathrm{Ag(s)}$$

Weak acids and bases ionise only partly, which is why equilibrium arrows are used:

$$\mathrm{HF(aq)} + \mathrm{H_2O(l)} \rightleftharpoons \mathrm{H_3O^{+}(aq)} + \mathrm{F^{-}(aq)}$$

$$\mathrm{HNO_2(aq)} + \mathrm{H_2O(l)} \rightleftharpoons \mathrm{H_3O^{+}(aq)} + \mathrm{NO_2^{-}(aq)}$$

$$\mathrm{CH_3COOH(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + CH_3COO^-(aq)}$$

$$\mathrm{HCN(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + CN^-(aq)}$$

$$\mathrm{HClO(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + ClO^-(aq)}$$

$$\mathrm{NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)}$$

$$\mathrm{CH_5N(aq) + H_2O(l) \rightleftharpoons CH_6N^+(aq) + OH^-(aq)}$$

Half-equations show the electrons that are gained or lost:

$$\mathrm{Cu^{2+}(aq) + 2\,e^- \longrightarrow Cu(s)}$$

$$\mathrm{Fe^{3+}(aq) + e^- \longrightarrow Fe^{2+}(aq)}$$

$$\mathrm{MnO_4^-(aq) + 8\,H^+(aq) + 5\,e^- \longrightarrow Mn^{2+}(aq) + 4\,H_2O(l)}$$

$$\mathrm{Zn(s) \longrightarrow Zn^{2+}(aq) + 2\,e^-}$$

$$\mathrm{2\,Cl^-(aq) \longrightarrow Cl_2(g) + 2\,e^-}$$

$$\mathrm{Al^{3+}(aq) + 3\,e^- \longrightarrow Al(s)}$$

$$\mathrm{O_2(g) + 4\,H^+(aq) + 4\,e^- \longrightarrow 2\,H_2O(l)}$$

$$\mathrm{Cr_2O_7^{2-}(aq) + 14\,H^+(aq) + 6\,e^- \longrightarrow 2\,Cr^{3+}(aq) + 7\,H_2O(l)}$$

Heating can break a compound down into simpler substances:

$$\mathrm{CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2(g)}$$

$$\mathrm{2\,KClO_3(s) \xrightarrow{\Delta} 2\,KCl(s) + 3\,O_2(g)}$$

$$\mathrm{2\,H_2O_2(aq) \longrightarrow 2\,H_2O(l) + O_2(g)}$$

$$\mathrm{2\,NaHCO_3(s) \xrightarrow{\Delta} Na_2CO_3(s) + H_2O(g) + CO_2(g)}$$

Many gas-phase reactions reach a dynamic equilibrium instead of going to completion:

$$\mathrm{N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)}$$

$$\mathrm{N_2O_4(g) \rightleftharpoons 2\,NO_2(g)}$$

$$\mathrm{2\,SO_2(g) + O_2(g) \rightleftharpoons 2\,SO_3(g)}$$

$$\mathrm{H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)}$$

Nuclear Equations

In nuclear notation the mass number is a superscript and the atomic number a subscript, both before the symbol. The nuclide ${}^{14}_{6}\mathrm{C}$ has 6 protons and 8 neutrons. The nuclide ${}^{238}_{92}\mathrm{U}$ has 92 protons and 146 neutrons. The nuclide ${}^{60}_{27}\mathrm{Co}$ has 27 protons and 33 neutrons. The nuclide ${}^{131}_{53}\mathrm{I}$ has 53 protons and 78 neutrons. The nuclide ${}^{56}_{26}\mathrm{Fe}$ has 26 protons and 30 neutrons. The nuclide ${}^{16}_{8}\mathrm{O}$ has 8 protons and 8 neutrons. Alpha decay of U-238:

$$\mathrm{{}^{238}_{92}U \longrightarrow {}^{234}_{90}Th + {}^{4}_{2}He}$$

Alpha decay of Ra-226:

$$\mathrm{{}^{226}_{88}Ra \longrightarrow {}^{222}_{86}Rn + {}^{4}_{2}He}$$

Alpha decay of Po-210:

$$\mathrm{{}^{210}_{84}Po \longrightarrow {}^{206}_{82}Pb + {}^{4}_{2}He}$$

Alpha decay of Pu-239:

$$\mathrm{{}^{239}_{94}Pu \longrightarrow {}^{235}_{92}U + {}^{4}_{2}He}$$

Alpha decay of Rn-222:

$$\mathrm{{}^{222}_{86}Rn \longrightarrow {}^{218}_{84}Po + {}^{4}_{2}He}$$

Alpha decay of Th-232:

$$\mathrm{{}^{232}_{90}Th \longrightarrow {}^{228}_{88}Ra + {}^{4}_{2}He}$$

Alpha decay of Am-241:

$$\mathrm{{}^{241}_{95}Am \longrightarrow {}^{237}_{93}Np + {}^{4}_{2}He}$$

Beta decay of C-14:

$$\mathrm{{}^{14}_{6}C \longrightarrow {}^{14}_{7}N + {}^{0}_{-1}e + \bar{\nu}_e}$$

Beta decay of H-3:

$$\mathrm{{}^{3}_{1}H \longrightarrow {}^{3}_{2}He + {}^{0}_{-1}e + \bar{\nu}_e}$$

Beta decay of Co-60:

$$\mathrm{{}^{60}_{27}Co \longrightarrow {}^{60}_{28}Ni + {}^{0}_{-1}e + \bar{\nu}_e}$$

Beta decay of Sr-90:

$$\mathrm{{}^{90}_{38}Sr \longrightarrow {}^{90}_{39}Y + {}^{0}_{-1}e + \bar{\nu}_e}$$

Beta decay of I-131:

$$\mathrm{{}^{131}_{53}I \longrightarrow {}^{131}_{54}Xe + {}^{0}_{-1}e + \bar{\nu}_e}$$

Beta decay of Cs-137:

$$\mathrm{{}^{137}_{55}Cs \longrightarrow {}^{137}_{56}Ba + {}^{0}_{-1}e + \bar{\nu}_e}$$

Beta decay of P-32:

$$^{32}_{15}\mathrm{P} \longrightarrow {}^{32}_{16}\mathrm{S} + {}^{0}_{-1}\mathrm{e} + \bar{\nu}_e$$

Beta decay of Pb-210:

$$^{210}_{82}\mathrm{Pb} \longrightarrow {}^{210}_{83}\mathrm{Bi} + {}^{0}_{-1}\mathrm{e} + \bar{\nu}_e$$

Positron emission by C-11:

$$^{11}_{6}\mathrm{C} \longrightarrow {}^{11}_{5}\mathrm{B} + {}^{0}_{1}\mathrm{e} + \nu_e$$

Positron emission by N-13:

$$^{13}_{7}\mathrm{N} \longrightarrow {}^{13}_{6}\mathrm{C} + {}^{0}_{1}\mathrm{e} + \nu_e$$

Positron emission by O-15:

$$^{15}_{8}\mathrm{O} \longrightarrow {}^{15}_{7}\mathrm{N} + {}^{0}_{1}\mathrm{e} + \nu_e$$

Positron emission by F-18:

$$^{18}_{9}\mathrm{F} \longrightarrow {}^{18}_{8}\mathrm{O} + {}^{0}_{1}\mathrm{e} + \nu_e$$

Positron emission by Na-22:

$$^{22}_{11}\mathrm{Na} \longrightarrow {}^{22}_{10}\mathrm{Ne} + {}^{0}_{1}\mathrm{e} + \nu_e$$

Electron capture in Be-7:

$$^{7}_{4}\mathrm{Be} + {}^{0}_{-1}\mathrm{e} \longrightarrow {}^{7}_{3}\mathrm{Li} + \nu_e$$

Electron capture in Ar-37:

$$^{37}_{18}\mathrm{Ar} + {}^{0}_{-1}\mathrm{e} \longrightarrow {}^{37}_{17}\mathrm{Cl} + \nu_e$$

Electron capture in Fe-55:

$$^{55}_{26}\mathrm{Fe} + {}^{0}_{-1}\mathrm{e} \longrightarrow {}^{55}_{25}\mathrm{Mn} + \nu_e$$

Electron capture in Cr-51:

$$^{51}_{24}\mathrm{Cr} + {}^{0}_{-1}\mathrm{e} \longrightarrow {}^{51}_{23}\mathrm{V} + \nu_e$$

Gamma emission from an excited Ni-60 nucleus:

$$^{60}_{28}\mathrm{Ni}^{*} \longrightarrow {}^{60}_{28}\mathrm{Ni} + \gamma$$

Gamma emission from an excited Sr-87 nucleus:

$$^{87}_{38}\mathrm{Sr}^{*} \longrightarrow {}^{87}_{38}\mathrm{Sr} + \gamma$$

Fission of uranium-235 after neutron capture:

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \longrightarrow {}^{141}_{56}\mathrm{Ba} + {}^{92}_{36}\mathrm{Kr} + 3\,{}^{1}_{0}\mathrm{n}$$

Deuterium and tritium fuse:

$$^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \longrightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n}$$

Nitrogen bombarded with alpha particles gives oxygen:

$$^{14}_{7}\mathrm{N} + {}^{4}_{2}\mathrm{He} \longrightarrow {}^{17}_{8}\mathrm{O} + {}^{1}_{1}\mathrm{H}$$

The first artificial radioactive isotope was made from aluminium:

$$^{27}_{13}\mathrm{Al} + {}^{4}_{2}\mathrm{He} \longrightarrow {}^{30}_{15}\mathrm{P} + {}^{1}_{0}\mathrm{n}$$

Measurements and Units

Quantities in the laboratory carry units, and many of those units have powers. At sea level the density of air is about 1.2 kg $\mathrm{m^{-3}}$, and a barometer reading corresponds to a pressure of 1.1 times 10^5 $\mathrm{N/m^2}$. The solution contains 0.1 mol $\mathrm{dm^{-3}}$ of hydrochloric acid, so 250 $\mathrm{cm^3}$ of it holds 0.025 mol. The reservoir holds 2.5 $\mathrm{km^3}$ of water spread over 150 $\mathrm{km^2}$. Near the surface of the earth, free fall has an acceleration of 9.81 m $\mathrm{s^{-2}}$.

The intensity of sunlight at the ground is roughly 950 W $\mathrm{m^{-2}}$ on a clear day. The density of ice is 0.92 $\mathrm{g/cm^3}$, so a block of volume 60 $\mathrm{cm^3}$ has a mass of 55 g. The pump delivers 0.8 $\mathrm{m^3/s}$, which is 800 $\mathrm{dm^3}$ every second. The specific heat capacity of aluminium is 0.9 J $\mathrm{g^{-1}}$ $\mathrm{K^{-1}}$.

The plate has an area of 0.5 $\mathrm{m^2}$ and a thickness of 10 cm, so its volume is 0.05 $\mathrm{m^3}$. A steel wire with a cross-section of 10.0 $\mathrm{mm^2}$ carries a load of 900 N, which is a stress of 90 $\mathrm{N/mm^2}$. The measured viscosity of the ice melt is 0.5 kg $\mathrm{m^{-1}}$ $\mathrm{s^{-1}}$. A cart accelerates at 4.5 $\mathrm{m/s^2}$ for 6 s and reaches a speed of 27 m/s.

The plate has an area of 2.0 $\mathrm{m^2}$ and a thickness of 2 cm, so its volume is 0.04 $\mathrm{m^3}$. The pump delivers 0.4 $\mathrm{m^3/s}$, which is 400 $\mathrm{dm^3}$ every second. The measured viscosity of the aluminium melt is 0.5 kg $\mathrm{m^{-1}}$ $\mathrm{s^{-1}}$. The specific heat capacity of aluminium is 0.9 J $\mathrm{g^{-1}}$ $\mathrm{K^{-1}}$.

A cart accelerates at 3.0 $\mathrm{m/s^2}$ for 4 s and reaches a speed of 12 m/s. The solution contains 2.0 mol $\mathrm{dm^{-3}}$ of sodium chloride, so 250 $\mathrm{cm^3}$ of it holds 0.5 mol. Near the surface of the earth, free fall has an acceleration of 9.81 m $\mathrm{s^{-2}}$. The density of ice is 0.92 $\mathrm{g/cm^3}$, so a block of volume 60 $\mathrm{cm^3}$ has a mass of 55 g.

The intensity of sunlight at the ground is roughly 1020 W $\mathrm{m^{-2}}$ on a clear day. The reservoir holds 6.0 $\mathrm{km^3}$ of water spread over 150 $\mathrm{km^2}$. A steel wire with a cross-section of 2.0 $\mathrm{mm^2}$ carries a load of 900 N, which is a stress of 450 $\mathrm{N/mm^2}$. At sea level the density of air is about 1.2 kg $\mathrm{m^{-3}}$, and a barometer reading corresponds to a pressure of 2.5 times 10^5 $\mathrm{N/m^2}$.

The specific heat capacity of copper is 0.39 J $\mathrm{g^{-1}}$ $\mathrm{K^{-1}}$. At sea level the density of air is about 1.2 kg $\mathrm{m^{-3}}$, and a barometer reading corresponds to a pressure of 1.2 times 10^5 $\mathrm{N/m^2}$. Near the surface of the earth, free fall has an acceleration of 9.8 m $\mathrm{s^{-2}}$. The solution contains 0.1 mol $\mathrm{dm^{-3}}$ of sodium chloride, so 250 $\mathrm{cm^3}$ of it holds 0.025 mol.

A steel wire with a cross-section of 8.0 $\mathrm{mm^2}$ carries a load of 900 N, which is a stress of 112.5 $\mathrm{N/mm^2}$. A cart accelerates at 2.0 $\mathrm{m/s^2}$ for 6 s and reaches a speed of 12 m/s.

Atoms and Nuclei in Running Text

The mass number of a nuclide stands before its symbol, as a raised number. The nucleus $^{14}\mathrm{C}$ decays by beta emission, and its half-life $t_{1/2}$ is 5730 years. Cobalt-60, written $^{60}\mathrm{Co}$, contains 27 protons and 33 neutrons. The isotopes $^{131}\mathrm{I}$ and $^{129}\mathrm{I}$ of iodine have the same chemical behaviour but different masses. A sample of $^{137}\mathrm{Cs}$ loses half of its atoms every 30.2 years, so the activity falls in a regular pattern.

The nucleus $^{226}\mathrm{Ra}$ decays by alpha emission, and its half-life $t_{1/2}$ is 1600 years. Polonium-210, written $^{210}\mathrm{Po}$, contains 84 protons and 126 neutrons. The isotopes $^{238}\mathrm{U}$ and $^{239}\mathrm{U}$ of uranium have

the same chemical behaviour but different masses. A sample of ^{241}Am loses half of its atoms every 432 years, so the activity falls in a regular pattern.

The nucleus ^{32}P decays by beta emission, and its half-life $t_{1/2}$ is 14.3 days. Hydrogen-3, written ^{3}H, contains 1 proton and 2 neutrons. The isotopes ^{222}Rn and ^{221}Rn of radon have the same chemical behaviour but different masses. A sample of ^{90}Sr loses half of its atoms every 28.8 years, so the activity falls in a regular pattern.

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Other Scripts in Running Text

Indices, powers of ten and ordinal endings also appear inside sentences. The kinetic energy E_k of the cart is 6 J at the bottom of the ramp. Expanding $(a + b)^2$ gives $a^2 + 2ab + b^2$. A tolerance of 10^{-3} is enough for the 12$^{\text{th}}$ iteration of the method. The roots of the quadratic are $x_1 = 2$ and $x_2 = 5$.

The coefficient of determination R^2 equals 0.84 for this fit. Each term satisfies $a_{n+1} = a_n + 7$, so with $a_1 = 2$ the 5$^{\text{th}}$ term is 30. The null hypothesis H_0 is rejected when the p-value is below 0.05. The 1$^{\text{st}}$, 2$^{\text{nd}}$ and 3$^{\text{rd}}$ terms are 5, 9 and 7.

The point $(x_1,\ y_1)$ lies on the line, and so does $(x_2,\ y_2)$. The square of the standard deviation, s^2, is called the variance. The half-life $t_{1/2}$ of the sample is 8 hours. Taking logarithms to base ten, $\log_{10} 3000$ is about 3.48.

The mass of the sample is 1 times 10^{-3} kg. The initial velocity v_0 is 7 m/s and the final velocity v_f is 10 m/s. The coefficient of determination R^2 equals 0.91 for this fit. The kinetic energy E_k of the cart is 15 J at the bottom of the ramp.

Each term satisfies $a_{n+1} = a_n + 5$, so with $a_1 = 3$ the 5$^{\text{th}}$ term is 23. The null hypothesis H_0 is rejected when the p-value is below 0.05. The roots of the quadratic are $x_1 = 5$ and $x_2 = 3$. The half-life $t_{1/2}$ of the sample is 8 hours.

The square of the standard deviation, s^2, is called the variance. The 1$^{\text{st}}$, 2$^{\text{nd}}$ and 3$^{\text{rd}}$ terms are 7, 2 and 9. A tolerance of 10^{-3} is enough for the 17$^{\text{th}}$ iteration of the method. The point $(x_1,\ y_1)$ lies on the line, and so does $(x_2,\ y_2)$.

Expanding $(a + b)^2$ gives $a^2 + 2ab + b^2$. The initial velocity v_0 is 7 m/s and the final velocity v_f is 10 m/s. The mass of the sample is 1.5 times 10^{-3} kg. Taking logarithms to base ten, $\log_{10} 3000$ is about 3.48.

Each term satisfies $a_{n+1} = a_n + 3$, so with $a_1 = 7$ the 5^{th} term is 19. The kinetic energy E_k of the cart is 21 J at the bottom of the ramp. The 1^{st}, 2^{nd} and 3^{rd} terms are 7, 5 and 3. The coefficient of determination R^2 equals 0.91 for this fit.

Taking logarithms to base ten, $\log_{10} 200$ is about 2.30. The point $(x_1,\ y_1)$ lies on the line, and so does $(x_2,\ y_2)$.